

Olist Customer Retention Analysis

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Tools: MySQL, Python (pandas, SQLAlchemy), Power BI

Dataset: Olist Brazilian E-Commerce Public Dataset , 99,441 orders and 93,358 unique customers

Business Question

What separates one-time buyers from repeat customers, and where should Olist focus its retention efforts?

The Short Answer

Customer retention was very low during the period covered by the dataset. About 97% of customers placed only one observed order, while 2,801 customers made more than one purchase.

Repeat customers were more valuable over the observed period, spending an average of R\$308.53 per customer compared with R\$160.74 for one-time buyers. However, product categories and review scores were very similar between the two groups, so these factors alone do not explain the difference in repeat purchasing.

Among customers who returned, the average time between the first and second purchase was 81 days. The analysis also found large differences in repeat purchase rates across sellers. These findings suggest two areas worth testing: the timing of customer re-engagement and the practices of sellers with unusually high repeat rates.

Key Findings

#	Question	Finding
1	How many customers purchased more than once?	About 97% placed only one observed order. Only 2,801 of 93,358 customers made more than one purchase during the dataset period.
2	Are repeat customers more valuable?	Yes. Repeat customers spent an average of R\$308.53 per customer compared with R\$160.74 for one-time buyers over the observed period.
3	Do they buy different products?	Not significantly. Four of the top five product categories overlapped between the two groups.
4	Are review scores different?	Very little. One-time buyers averaged 4.15 compared with 4.21 for repeat customers.

5	When do repeat customers return?	Customers who made a second purchase returned after an average of 81.2 days. Later purchase gaps averaged about 62–63 days.
6	Do some sellers have higher repeat rates?	Yes. The highest observed seller repeat rate was 22.73%, compared with a platform average of 2.76%. The available analysis did not identify location, category mix, or review score as clear explanations for the difference.

Recommendations

1. Test a customer reactivation campaign before the average second-purchase point

Customers who returned made their second purchase after an average of 81 days. Olist could test re-engagement messages before this point, for example between days 55 and 65 after the first delivered order.

This timing should be treated as a starting point for an experiment rather than a proven optimal window. Results should be measured against a control group and adjusted based on conversion.

2. Study sellers with unusually high repeat rates

Some sellers recorded repeat purchase rates several times higher than the platform average. Olist should compare these sellers with lower-retention sellers to identify differences that were not available in the current analysis.

Areas worth investigating include delivery performance, packaging, communication, pricing, promotions, and post-purchase follow-up.

3. Investigate additional causes of low repeat purchasing

Similar review scores and category preferences suggest that these two factors alone do not explain the retention gap. Further analysis could examine delivery times, cancellations, refunds, pricing, customer acquisition sources, and post-purchase communication.

4. Measure retention by customer cohort

The current analysis covers customers observed within a fixed dataset period. Future retention analysis should group customers by first-purchase month and give each cohort a consistent observation window. This would provide a clearer view of how repeat purchasing changes over time.

Investigation

Query 1, Order Frequency Distribution

The first query measured how often customers purchased during the dataset period.

About 97% of the 93,358 customers placed only one observed order. Only 2,801 customers made more than one purchase, and 181 customers placed three orders.

This established the scale of the retention problem and became the starting point for the rest of the analysis.

Query 2, Revenue by Customer Type

I compared one-time and repeat customers to understand their contribution to revenue.

One-time buyers generated approximately R\$14.56M, or 94.4% of observed revenue.

Repeat customers generated approximately R\$864K, or 5.6%.

Although repeat customers represented a small share of the customer base, their average observed spend was higher: R\$308.53 per customer compared with R\$160.74 for one-time buyers.

Query 3, Top Categories by Customer Type

I compared the most frequently purchased product categories for one-time and repeat customers.

Four of the top five categories overlapped between the two groups, including health and beauty, home and bath, sports and leisure, and computer accessories.

This suggests that category preference alone does not explain why some customers purchased again.

Query 4, Review Scores by Customer Type

I compared average review scores to test whether lower customer satisfaction was associated with one-time purchasing.

One-time buyers had an average review score of 4.15, compared with 4.21 for repeat customers.

The small difference suggests that review score alone does not explain the large difference in repeat purchasing.

Query 5, Days Between Purchases

I used window functions to measure the time between consecutive purchases for repeat customers.

The average gap between the first and second purchase was 81.2 days. The average gap for later purchases fell to approximately 62–63 days.

This suggests that earning the second purchase may be an important stage in the customer relationship and provides a useful starting point for testing the timing of re-engagement campaigns.

Query 6, Seller-Level Retention Analysis

The final query compared repeat purchase rates across sellers.

The highest observed seller repeat rate was 22.73%, compared with a platform average of 2.76%. I compared seller location, category mix, and review scores, but these factors did not clearly explain the difference.

The result raises a further business question: what are high-retention sellers doing differently? Answering that would require additional operational data.

Files

File	Description
olist_analysis.sql	Six SQL queries used in the analysis
Victoria_Johnson_Olist_Case_Study.docx	Full case study
Victoria_Johnson_Olist_Dashboard.pdf	Power BI dashboard export
final_import.py	Python script used to import the CSV files into MySQL with explicit data types

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Tools and Techniques

- **MySQL:** Data preparation and analytical queries
- **Python (pandas, SQLAlchemy):** CSV-to-MySQL import, explicit data type mapping, and index creation
- **Power BI:** Interactive dashboard for customer frequency, revenue, category comparison, and purchase timing
- **SQL:** CTEs, window functions including RANK, ROW_NUMBER and LAG, multi-table joins, CASE WHEN segmentation, DATEDIFF, and conditional aggregation

Data Engineering Notes

The data import created a separate technical challenge during the project.

The problem

The initial pandas import created ID and text columns as TEXT fields. This prevented normal indexing of the join columns and contributed to slow queries and database connection errors.

The main errors were:

- Error 1170: BLOB/TEXT column used in key specification without a key length
- Error 2013: Lost connection to MySQL server during query

The fix

I re-imported the tables using explicit data types, including VARCHAR fields for IDs and appropriate numeric types for prices and counts.

I then created 14 indexes across the main join columns. A category analysis query that previously timed out completed in 8.13 seconds after the schema and indexing changes.

What I learned

This part of the project showed me that analysis is not only about writing the final SQL query. Data types, indexing, table structure, and query performance can determine whether an analysis works at all.

About Me

Victoria Johnson is a Business and Data Analyst with five years of experience in US remote B2B sales in the building materials and manufacturing industry. My work focuses on using data to investigate business problems, understand customer behavior, and support better commercial decisions.

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